or, equivalently

$$r(X,Y)=\frac{n\sum_{i=1}^{n}x_iy_i-\sum_{i=1}^{n}x_i\sum_{i=1}^{n}y_i}{\sqrt{n\sum_{i=1}^{n}x_i^2-\left(\sum_{i=1}^{n}x_i\right)^2}\sqrt{n\sum_{i=1}^{n}y_i^2-\left(\sum_{i=1}^{n}y_i\right)^2}}$$

### 4.4.2 Properties

1. The correlation coefficient between $X$ and $Y$ is same as the correlation coefficient between $Y$ and $X$ (*i.e,* $r_{xy} = r_{yx}$).
2. The correlation coefficient is free from the units of measurements of $X$ and $Y$
3. The correlation coefficient is unaffected by change of scale and origin.

   Thus, if $u_i = \frac{x_i - A}{c}$ and $v_i = \frac{y_i - B}{d}$ with $c \neq 0$ and $d \neq 0$ $\quad i$=1,2, ..., $n$

$$r=\frac{n\sum_{i=1}^{n}u_iv_i-\sum_{i=1}^{n}u_i\sum_{i=1}^{n}v_i}{\sqrt{n\sum_{i=1}^{n}u_i^2-\left(\sum_{i=1}^{n}u_i\right)^2}\sqrt{n\sum_{i=1}^{n}v_i^2-\left(\sum_{i=1}^{n}v_i\right)^2}}$$

   where $A$ and $B$ are arbitrary values.

**Remark 1:** If the widths between the values of the variabls are not equal then take $c = 1$ and $d = 1$.

#### Interpretation

The correlation coefficient lies between -1 and +1. *i.e.* $-1 \leq r \leq 1$

- A positive value of '$r$' indicates positive correlation.
- A negative value of '$r$' indicates negative correlation
- If $r = +1$, then the correlation is perfect positive
- If $r = -1$, then the correlation is perfect negative.
- If $r = 0$, then the variables are uncorrelated.
- If $|r| \geq 0.7$ then the correlation will be of higher degree. In interpretation we use the adjective 'highly'
- If $X$ and $Y$ are independent, then $r_{xy} = 0$. However the converse need not be true.

#### Example 4.1

The following data gives the heights(in inches) of father and his eldest son. Compute the correlation coefficient between the heights of fathers and sons using Karl Pearson's method.

| *Height of father* | 65 | 66 | 67 | 67 | 68 | 69 | 70 | 72 |
|---|---|---|---|---|---|---|---|---|
| *Height of son* | 67 | 68 | 65 | 68 | 72 | 72 | 69 | 71 |



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